

Use of Handheld Thermal Imager Data for Airborne Mapping of Fire Radiative Power and Energy and Flame Front Rate of Spread

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Abstract—Infrared (IR) remote sensing is increasingly used in studies of vegetation fire behavior, and high spatiotemporal resolution investigations often require data to be collected from airborne platforms, for example, standard helicopters. This paper aims to extend the range of conditions under which low-cost “handheld” thermal imaging cameras can be employed in such studies, particularly by enabling the effective and efficient geometric correction of thermal imagery collected from such devices, even when viewing far off-nadir (e.g., out of a side door or window). The approach is based on the automated detection of a set of fixed thermal “ground control points,” coupled with the use of a linear transformation matrix for warping the raw IR imagery to a fixed coordinate system. The output set of geometrically corrected brightness temperature and radiance images can be used to derive fire radiative power (FRP) and flame front rate of spread (ROS). We demonstrate and test our IR image processing methods on a series of case study fires, ranging from a small-scale laboratory to a 945-m² outdoor experimental burn. We compare mapped information on FRP obtained from simultaneous nadir and off-nadir views, where we find differences that are in part controlled by flame structure and/or view angle. In the large open fire case, we compare the mapped fire radiative energy and ROS to simultaneously acquired aerial photography that provides the position of fuel and flames in high detail, and we demonstrate how these data sets can be used to explore various aspects of fire behavior.

Index Terms—Fire radiative energy (FRE), fire radiative power (FRP), georeferencing algorithm, rate of spread (ROS), thermal imagery.

I. INTRODUCTION

ALTHOUGH most research related to individual fire front rate of spread (ROS) and/or fire energy release rate still utilizes long-established point-based measurement approaches (e.g., [1]–[3]), the use of thermal imaging systems to study propagating fires is increasingly widespread, including from

airborne platforms (e.g., [4]–[6] and [7]–[14]). However, the airborne thermal imagers typically used so far have mostly been mounted in dedicated survey aircraft, which can then use automated geocorrection procedures that require the position of the sensor in the aircraft to be tightly controlled, along with a suitably accurate model of the sensor optics and terrain relief [15]. By contrast, small thermal imaging cameras are increasingly available at relatively low cost and can be operated opportunistically either mounted or handheld from a fixed or moving elevated position (e.g., a high platform or standard helicopter) [16]. However, image georeferencing remains a limitation with such systems, particularly where the camera is mounted on a mobile platform (e.g., when operated from a helicopter or other moving platform under open fire conditions) and/or when viewing from far off-nadir (e.g., an elevated viewing platform). In many studies, these situations have been avoided, and the measurement geometry was set in order to simplify the georeferencing process. For example, McRae *et al.* [17] used a helicopter equipped with a vertical-viewing open floor port that allowed a thermal camera to collect nadir-view data from directly above the fire. Conversely, Pastor *et al.* [18] used an off-nadir viewing camera, but in a fixed position within a laboratory setup. Here, we demonstrate methods to extend the range of conditions under which these types of small widely available “handheld” thermal imaging cameras can be effectively employed in studies of fire behavior. In particular, we develop and test a georeferencing approach that can be applied to thermal imagery data of propagating fires, including at the sites of large outdoor experimental or prescribed fires, which are the most representative of “natural” wildfire-type conditions (e.g., [19]). Our aim is to demonstrate an approach that can accurately and semiautomatically georeference thermal imagery obtained by a handheld thermal camera operated by simply pointing it out of the door or window of a standard helicopter, which is the most practical and low cost measurement in such a situation. We use these data to accurately map spatiotemporal variations in flame front ROS and fire radiative power (FRP) on a series of experimental fires, including a 945-m² open vegetation fire conducted in Northumberland, U.K. To differentiate between measurements at an individual location and those spatially integrated across the fire scene, we define FRP_p as the radiative power output of a single fire pixel (expressed in watts) and FRP as the sum of all the FRP_p measures made instantaneously at all fire pixels detected within the burning plot. The temporal integrals of these two measures represent the fire radiative

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energy (FRE) emitted during the complete combustion process, both at a single pixel and across the entire plot (i.e., FRE_p and FRE).

II. BACKGROUND OF THE APPROACH

The geometric correction approach developed here is based on the direct linear transformation (DLT) method [20], whose use for calculating ROS under laboratory-scale fire conditions was described in [18]. Without knowing the camera position, or having a camera calibration, the method can map a raw thermal image to a new imaging grid (e.g., the view as would be seen from nadir) using a set of static ground control points (GCPs). Several other methods of image transformation are available. The choice of the DLT algorithm was motivated by (i) its capability to handle varying pixel sizes within the image (e.g. unlike an affine transformation) and (ii) its requirement for few GCPs (e.g. unlike polynomial warping). We apply the DLT method to imagery collected by an AGEMA 550 middle infrared (MIR) thermal imaging camera, whose measurements can be used to derive FRP and FRE from small- to large-scale fires [12], [21], [22]. With the standard lens attached, the AGEMA 550 measures radiation across a $20^\circ \times 15^\circ$ field of view (FOV) in a narrow waveband centered on $\lambda = 3.9 \mu\text{m}$, and when viewing fires, it is usually set to record repetitive images at around 0.25 Hz over an elevated brightness temperature (BT) range of 470 K–900 K. This avoids pixel saturation over even intensely burning areas, but it does prevent a signal from being measured over the ambient (≈ 300 K) background, so nonfire areas essentially appear blank. As a result, the GCPs required by the DLT algorithm need to be formed from small “bonfires” or other fixed hot targets that can remain hot for the entire fire duration and which are normally placed just outside the extremities of the burning plot. To facilitate the identification of these “thermal” GCPs, an algorithm based on the subwindow spatial correlation of two temporally neighboring thermal images was developed. This algorithm automatically locates the set of fixed thermal GCPs located close to the four corners of the (usually semirectangular) experimental fire plot. The DLT and control point detection algorithms together form a powerful tool used to georegister the AGEMA 550 thermal imagery and thus produce a geolocated map of fire pixel BT and FRP from each imaging frame. These maps can then be used to generate other products, for example, information on ROS, FRP, and FRE, as described in the following sections.

III. METHODOLOGY

Our overall geometric correction approach can be summarized in the following steps (the different angles and dimensions used hereinafter are shown in Fig. 1).

- 1) Assume that the burning plot has a characteristic length $L[m]$, aligned with the longest axis of the camera frame of length of n pixels. The camera FOV is α rad, so its instantaneous FOV is $\alpha_{\text{ifov}} = \alpha/n$ [rad]. If the burn plot is a square with, for example, $L = 100$ m, then the fire duration is likely to be on the order of many minutes,

which will allow tens to hundreds of thermal imaging frames to be collected.

- 2) The flight altitude of the helicopter ($h[m]$) is defined such that the full plot remains in the FOV during the entire fire duration, $h = (3/4L)/\tan(\alpha/2)$.
- 3) Prior to the ignition of the main plot, four small “GCP” fires are ignited close to the plot corners, and their accurate geographic coordinates are recorded using differential GPS. Their size must be larger than the average pixel size (APS) $\bar{p}$ defined by the view zenith angle θ and the distance of the helicopter to the center of the plot $d_h = h/\sin \theta$ since $\bar{p} = 2d_h \tan(\alpha_{\text{ifov}}/2)/\cos \theta$. Hovering helicopters cannot remain fully static and often drift and rotate somewhat. This is taken into account in the algorithm, and the only limitation is that images where part of the plot and/or one or more of the four GCP fires fall outside of the instrument FOV are removed by screening the time series prior to further processing.
- 4) The postprocessing of each screened thermal image starts with the retrieval of the GCP fire pixel coordinates, which is based on the spatial autocorrelation between temporally neighboring scenes.
- 5) Using the GCP fire pixel and geographic coordinates, the DLT algorithm is used to warp each thermal image to a new geometry that is designed to replicate a nadir view, with an output spatial resolution selected to be slightly finer (by around 10%) than the estimated mean pixel size of the raw thermal imagery, $\bar{p}$ as defined in Step 3). The spatial sampling is conducted following the method in [18], where the BT or radiance of each output “warped” image pixel is interpolated from the four spatially closest pixels in the raw input image.
- 6) The resulting data set represents a time series of georeferenced thermal imagery with a predefined constant pixel size, from which maps of the fire front ROS and per-pixel map values of FRP (FRP_p) and FRE (FRE_p) can be calculated.

The autolocation of the GCPs [see Step 4] is a challenging task since the algorithm is iterative and can diverge from a solution quite easily. Pixels representing the thermal GCPs in the first image of the temporal sequence need to be manually located, and then, using the spatial correlation of small pixel windows centered around each of those locations, the algorithm identifies the best match locations in the subsequent image. This “best match” location is then used as the thermal GCP location for the following spatial correlation procedure, and so on. A series of thresholds is required in order to ensure that thermal GCPs are correctly identified. These thresholds consist of (i) strong spatial correlation between the two output warped images; (ii) a relatively constant number of pixels per thermal GCP; and (iii) low BTs outside the polygon formed by the four control points (noting that the GCP fires are not always located at exactly the plot corners). Images failing one or more of these tests, or which cause problems for some other reason, are removed from the time series.

Experience shows that the current autocontrol point selection and geometric warping algorithm operating on data from the

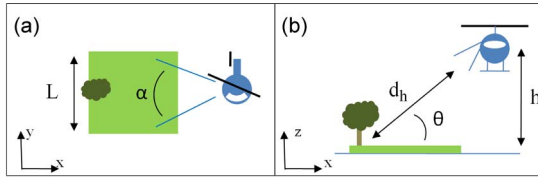


Fig. 1. Sketch representations of the field measurement approach described in Section III, whereby a thermal imaging camera operated from a hovering helicopter observes a rectangular plot on the ground. The various angular and distance terms are defined hereinafter.

AGEMA 550 MIR camera system can deliver a final geometric error of typically around two or three pixels (see Section IV). This could probably be improved further if more control point features were available, for example, from the ambient background. However, as already stated, when operated in the “fire measurement” range, the ambient regions of the 12-bit AGEMA 550 imagery typically contain no useful data. This is because the highly nonlinear relationship between radiance and BT at MIR wavelengths results in the spectral radiances emitted by a 1000 K fire and a 300 K background differing by a factor of around 5600 (assuming a $3.9\text{-}\mu\text{m}$ wavelength and blackbody behavior). Covering this extreme signal range linearly with a 12-bit imager such as the AGEMA 550 results in an unusable ≈ 30 K equivalent BT increment for each radiometric measurement (DN) increment at ambient temperatures. This BT increment quickly reduces as temperatures increase, but for this reason, most MIR thermal cameras divide up the potential temperature measurement range into subranges using different gain settings in order to improve the radiometric precision (e.g., the 473 K–1073 K “fire measurement” range used here). This results in us having to deploy “bonfires” or other hot targets such as thermal GCPs, rather than being able to use ambient background thermal features, thus limiting the GCP number. For long wave infrared (LWIR) thermal cameras, the more linear BT versus spectral radiance relationship [23] results in a much reduced signal range (the 300 K–1000 K spectral radiances differ by a factor of around 30), and a 12-bit $11\text{-}\mu\text{m}$ imager can therefore potentially cover the ambient to fire pixel temperature range with a single gain setting and still deliver a 0.5 K BT increment per DN at ambient temperatures. Unfortunately, LWIR measurements are far less suited to the derivation of FRP in cases where the fire under fills pixels [24], so applications aiming toward FRP and FRE assessment are best conducted with MIR wavelength imaging systems. The use of the new generation of MIR thermal imaging cameras that can collect near-simultaneous images at two dynamic range settings, which can then be combined to form a single image covering, e.g., an ambient to 1000 K range [26], would very likely help to improve the geometric warping performance since this would allow ambient background thermal features to act as GCPs. However, at present, these systems are currently very expensive compared to the more standard MIR camera used here and are beyond the reach of many fire research or management organizations.

The DLT algorithm (Step 5 of the methodology) used to perform the georeferencing contains some assumptions. We refer to [18] for a detailed mathematical description, but the

main limitation of the approach is the assumption that the burning plot is a flat plane. The method is based on a linear warping between the location of the control points within the image $X_I = (p_x, p_y)$ and their actual ground coordinates $X_R = (x, y)$

$$X_R = H X_I. \quad (1)$$

The plot can be sloping, but if it has significant changes in slope or other topography, then the transformation matrix H is nonlinear, and the DLT algorithm becomes increasingly invalid as the topography becomes more extreme. To avoid any dependence on the coordinate system, (1) has to be normalized, and a new scaling variable is required to close the system. Therefore, H is a 3×3 matrix with $X_I = (p_x, p_y, 1)$ and $X_R = (x, y, 1)$, and its calculation requires at least three GCPs. However, the thermal GCP coordinates determined by the spatial correlation procedure will very likely contain some error, so following [18], we use four GCPs to overdetermine (1). Another limitation of the DLT algorithm relates to radiation from the flame. Although the narrow-band $3.9\text{-}\mu\text{m}$ wavelength of the AGEMA 550 avoids absorption and emission features caused by flame gases (mainly CO_2 [21]), when the flame contains significant soot particles (e.g., as in the yellow flame of a candle), the thermal imagery observations can be somewhat dependent on the viewing angle [12]. In this situation, the DLT algorithm can introduce some distortion since the transformation matrix H is again not operating only between two planar surfaces. This issue is further addressed in Section IV and would be present in most generic warping schemes. In spite of these limitations, we believe the geometric correction approach developed here can be applied in an efficient and effective manner to derive spatial maps of fire pixel BT and radiance, which can then be used to calculate ROS, FRP, and FRE from helicopter-borne thermal imagery.

For comparison to the DLT approach, we also applied a simpler “Average Pixel Size (APS)” method to calculate plot-integrated FRP values that assumes a constant pixel size across each imaging frame, calculated (according to Step 3 mentioned earlier) from the camera to plot-center distance and the view zenith angle. These two measures are simple to evaluate using the helicopter GPS record and the ground coordinates of the GCPs. Since the APS method does not include adjustment for the horizontal pixel size variation across the image, it functions best when fire activity and view zenith angle variations across the scene are limited and are not able to provide spatially explicit (i.e., mapped) information. However, it is able to provide estimates of plot-integrated FRP and fire-integrated FRE for the observed fires. The following section details a series of case studies demonstrating the algorithm’s use on a series of increasingly more complex fire situations, culminating in a 945-m^2 burning plot imaged from an overhead helicopter platform.

IV. CASE STUDIES

Our case studies aimed to provide an evaluation of the overall geometric correction algorithm performance and applicability when used with the handheld AGEMA 550 $3.9\text{-}\mu\text{m}$ system described in [21]. There are three test cases, each examining

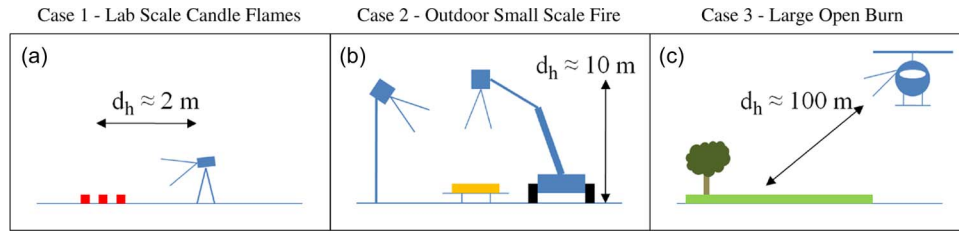


Fig. 2. Sketch representations of the three different test fire cases studied in Section IV, illustrating the target and thermal imaging camera locations (d_h represents the approximate distance between them).

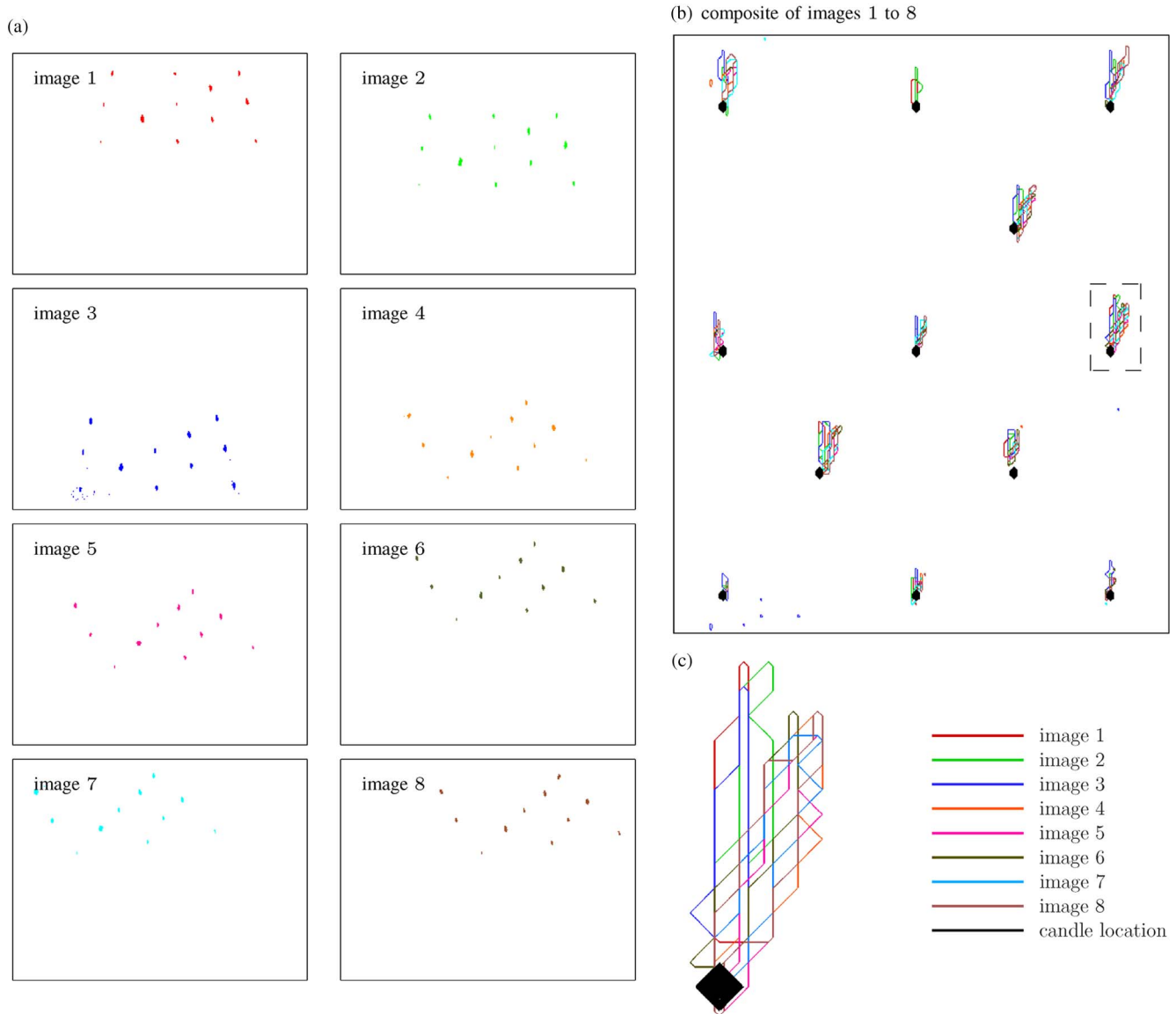


Fig. 3. Data from the indoor laboratory-scale fire situation (Case study 1). (a) Raw imagery taken by the AGEMA 550 thermal imaging camera and showing a rectangular shaped area ($44.5 \text{ cm} \times 57 \text{ cm}$) marked out with a series of burning candles and viewed by the camera as shown in Fig. 2(a). Imagery was taken off-nadir and with different azimuth and zenith viewing angles, varying over 0° to 15° and 17° to 25° , respectively. Depicted here are the isocontours of the pixel BTs exceeding 520 K, and the varying position of the candles between each imaging frame is clearly apparent. (b) Eight images shown in (a) warped to a single geometry using the methods described in Section III and then used to create this single composite image. The true location of each candle is shown by the black dot, while the $> 520 \text{ K}$ BT isocontour of each fire is shown in the various colors. (c) Zoom of panel (b) within the area of the dashed rectangle.

different types of fire scene at different spatial scales (see Fig. 2). The first and simplest is a small laboratory-scale ($0.45 \text{ m} \times 0.57 \text{ m}$) rectangular field of interest, whose corners are marked with candles and which is observed by the thermal camera from several positions. The second is an outdoor small-

scale ($1 \text{ m} \times 1 \text{ m}$) burning fuel bed, viewed simultaneously from two fixed cameras located at different positions with different viewing angles. Last is the open vegetation fire of size $45 \text{ m} \times 21 \text{ m}$, observed via a hovering helicopter at a view zenith angle of around 55° .

TABLE I
DESCRIPTION OF THE DIFFERENT SMALL-SCALE FIRES USED IN CASE STUDY 2 OF THIS WORK (SEE SECTION IV-B)

Fire	Ignition	Geometry $l \times w \times h$ (cm ³)	Mass (kg)	Comment
01	downwind	120x90x14	2.3	
02	upwind	120x90x14	2.25	
03	upwind	120x49x14	1.0	fuel placed on wire mesh
04	downwind	120x49x14	1.1	fuel placed on wire mesh
05	upwind	120x49x14	1.0	same as fire 03
06	upwind	120x90x14	1.8	similar to fire 02 with lower mass
07	upwind	120x49x14	1.0	similar to fire 03 but on the scale
08	upwind	120x49x28	2.1	similar to fire 07 but twice the mass

TABLE II
PERCENT DIFFERENCE IN FRE CALCULATED FROM THE THERMAL IMAGERY CAPTURED BY THE NADIR-VIEWING AGEMA 550 INFRARED CAMERA OF CASE STUDY 2, AS RETRIEVED USING THE DLT ALGORITHM (FRE_{DLT}^n) AND THE APS APPROACH (FRE_{APS}^n). THE CHARACTERISTICS OF EACH FIRE ARE LISTED IN TABLE I

Fire	01	02	03	04	05	06	07	08
$\frac{FRE_{DLT}^n - FRE_{APS}^n}{FRE_{APS}^n} \times 100$ (%)	-0.2	-0.09	-0.03	-0.04	-0.1	-0.1	-0.4	-0.01

A. Case Study 1—Laboratory-Scale Candle Flames

The main purpose of this case study was to evaluate the basic performance of the DLT algorithm component of the overall geometric correction scheme, when taking a set of multitemporal thermal images collected across a series of camera locations and viewing angles and warping them to a single fixed output geometry. For this test, eight infrared images were collected of a rectangular area (44.5 cm $\times$ 57 cm) that had each corner and several points on the diagonal marked out with burning candles that were easily detectable in the thermal images [see Fig. 3(a)]. The AGEMA 550 was mounted on a moving tripod such that all images were taken off-nadir from distances of $\sim$ 2 m to 2.5 m and with view zenith and azimuth angles of $\sim$ 17°–25° and 0°–15°, respectively. The view azimuth angle is defined here from the reference axis normal to the shortest edge of the plot. The fixed ground coordinates of each candle were carefully recorded, and these locations were used as the thermal GCPs for the DLT warping procedure. Since the aim of this test case was to evaluate the DLT component of the overall algorithm, to minimize other sources of error, we manually selected the GCP fire pixel locations rather than using the autolocation procedure.

A composite of the eight output georeferenced image frames is shown in Fig. 3(b). In each warped frame, the base of each candle flame is located within 2 or 3 pixels of the true location (equivalent to 1.0–1.5 cm), thus confirming that the DLT algorithm has correctly warped the set of thermal scenes to within an accuracy of a few pixels. The results, however, also highlight limitations with regard to the 3-D structure of a scene. Images 4, 5, 6, 7, and 8, which were taken at different lateral orientations, show tilted flames after warping (see Fig. 3(c); the isoline of BT $T > 520$ K is vertical for images 1, 2, and 3 but not for images 4, 5, 6, 7, and 8). However, this distortion appears only in cases when the pixel size is fine enough to resolve the flame structure. When imaging larger fires from much greater distances (e.g., via a helicopter), the image pixel size is usually similar to or even larger than the flame dimensions [22], and the geometry of the fire scene can then be well captured with the 2-D approximation in all but the most extreme fire situations

(e.g., high intensity crown fires where flame lengths may reach hundreds of meters).

B. Case Study 2—Outdoor Small-Scale Fire

The previous section demonstrated that the DLT algorithm can perform adequate georeferencing of thermal scenes in situations where the observed features can be considered to approximate 2-D objects. This second case study is focused on the use of the algorithm in the types of small-scale fire situations previously used to investigate relations between FRE and fuel consumption (e.g., [12] and [21]). To do this, we conducted a series of small-scale fires using 1 m² of straw as fuel, similar to those described in [21]. Two identical AGEMA 550 MIR cameras were used to record simultaneous imagery of each fire, one viewing from nadir (at a distance $h = 8.10$ m) and one viewing from off-nadir (view zenith angle $\theta = 40.7^\circ$ and viewing distance $h = 6.40$ m). This is somewhat similar to the conditions explored by Freeborn *et al.* [12] to derive relations between FRE and fuel consumption at different view angles, except that, in that study, the two cameras operated at different wavelength ranges (one LWIR [8–14 μ m] and one MIR [3–5 μ m]). We conducted eight fires in total, and their configuration and geometries are summarized in Table I. Vertical and horizontally viewing optical video records were also collected for each fire, and in this study, the thermal GCPs were defined by the fixed corners of the underlying (sand-covered) metal plate upon which the fuel was homogeneously spread, so as with Case study 1, the autocontrol point selection component of the overall geometric correction approach was not applied. As a first evaluation of the DLT algorithm's use in FRP assessment, Table II shows for each fire the difference between the final FRE evaluated by temporally integrating the FRP observations made by the nadir-viewing camera, both when the image pixel size is retrieved by the DLT algorithm (FRE_{DLT}^n) and by the simpler APS (FRE_{APS}^n) approach. In this particular configuration, where the nadir-viewing camera is fixed at a known location and the fuel bed is homogeneous, we can assume a single mean pixel size for all thermal pixels covering the 1-m² fire area.

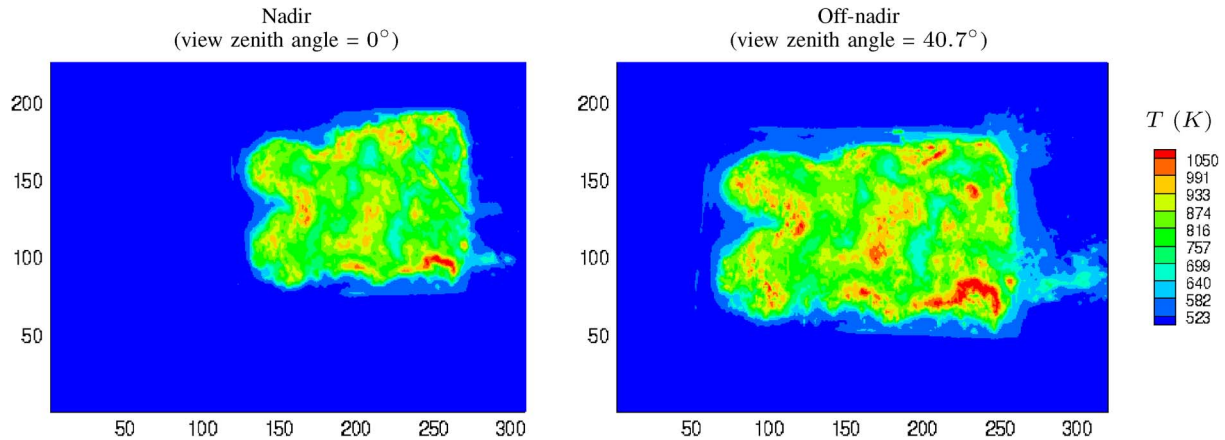


Fig. 4. Raw BT images from the outdoor small-scale fire situation (Case study 2), shown for Fire 02 at $t = 80$ s from both the nadir and off-nadir viewing AGEMA 550 thermal imaging cameras (which operate at $3.9 \mu\text{m}$). See Fig. 5 for comparison with the warped images. Axis labels represent the pixel indexes.

We assume here that the fire is well centered within the image frames such that we can neglect image distortion [24]. Under such conditions, the APS approach to FRE derivation is likely to be the most accurate since the pixel area is well estimated and the method does not rely on the selection of GCP locations that can introduce errors, as discussed in Section III. In this particular nadir-viewing situation, the differences between FRE_{DLT}^n and FRE_{APS}^n are small (a maximum of 0.4% as shown in Table II) and are solely due to geometric effects, probably related to GCP location uncertainties. These can be induced by changes in wind speed and direction that caused slight movements in the mast-mounted camera's position and which caused the true image coordinates of the GCPs to vary over time by a few pixels.

A second focus is on the comparison of the FRP measures derived from the data collected by the two AGEMA 550 cameras observing at different view zenith angles. One objective was to evaluate the effect of view angle on the computation of FRP. To address this, the thermal images of the two intercalibrated thermal cameras were georeferenced to the same base frame, and the FRP was derived using the MIR radiance method in [25] and [21]. Before comparing the FRP derived from the nadir and off-nadir views, it is useful to examine Fig. 4, which shows the raw BT images of Fire 02 as derived from both thermal imaging cameras at $t = 100$ s. Since the distance between the camera and the fire was slightly different for the nadir and off-nadir views, the APS of the raw images without any adjustment for the viewing angle also differs (i.e., 1 cm for the nadir and 0.75 cm for the off-nadir camera). To avoid loss of information when the high-spatial-resolution off-nadir images are warped and to facilitate interimage comparison, a common spatial resolution of 0.7 cm was set for all warped imagery [see Step 5] of Section III). Varying this output spatial resolution between 0.5 and 1 cm changed the total FRE measured by each camera by less than 0.03%. Thermal images of the flaming regime of Fires 02, 05, and 08 are shown in Fig. 5 (see Fig. 4 for a comparison to the corresponding raw images of Fire 02). A comparison of the warped imagery from the nadir and off-nadir views shows that the BT distribution for Fires 05 and 08 is more dependent on the camera viewing orientation than it is for Fire 02 (i.e., more variability in the shape and value of the

isocontours). A possible explanation comes from the different configurations of Fires 02, 05, and 08 (see Table I). In particular, Fires 05 and 08 experienced a stronger flaming regime than did Fire 02 since both have a larger surface area in contact with the ambient air and this enhances oxygen and fuel gas mixing. The fuel in Fire 05 was mounted atop a wire mesh that allowed air flow from underneath as well as from above the fuel bed, while Fire 08 had the same mass as Fire 02 but arranged in a deeper fuel bed. This resulted in Fires 05 and 08 having slightly taller flames, whose 3-D structure affects the output from the DLT algorithm as described in Section IV-A. However, even for Fire 02, we can see that the two warped images appear somewhat different, with higher temperatures toward the south of the off-nadir image. The off-nadir camera was positioned at location $(x = 1.5, y = -4)$ in the relative reference axis of each image shown in Fig. 5 and therefore had a better viewing angle with which to resolve the flame profile in the southern part of the fuel bed (i.e., at locations with low y values in the axes of Fig. 5). As temperatures are typically higher closer to the fuel bed (entrainment produced by the flame tends to cool the local temperature within the plume vertical profile) and as there is less material between the flame base and the camera (e.g., less cooler soot that may absorb some of the thermally emitted radiation), the off-nadir camera records higher temperatures toward the southern part of the image. To understand in detail how the viewing angle affects the FRP measurements taken here, two effects must be addressed: 1) the distortion of the flame 3-D geometry during geometric correction and 2) the fact that the two cameras are not looking at exactly the same fire scene (e.g., the effect of radiation absorption by soot particles located in the flame and smoke, and whose path length varies with view angle). Detailed consideration of these points probably requires the use of numerical fire simulations in addition to observations, such as those in [26] and [27] (or the preliminary work in [28]). However, to emphasize the potential impact of the view angle on FRP measures, Fig. 6 shows the difference between the time-integrated FRP (i.e., the FRE) derived from the geometrically warped nadir and off-nadir views of each fire. Fires having a configuration more favorable to a stronger flaming regime (e.g., Fires 03, 04, 05, and 08 of Fig. 5) typically show a larger

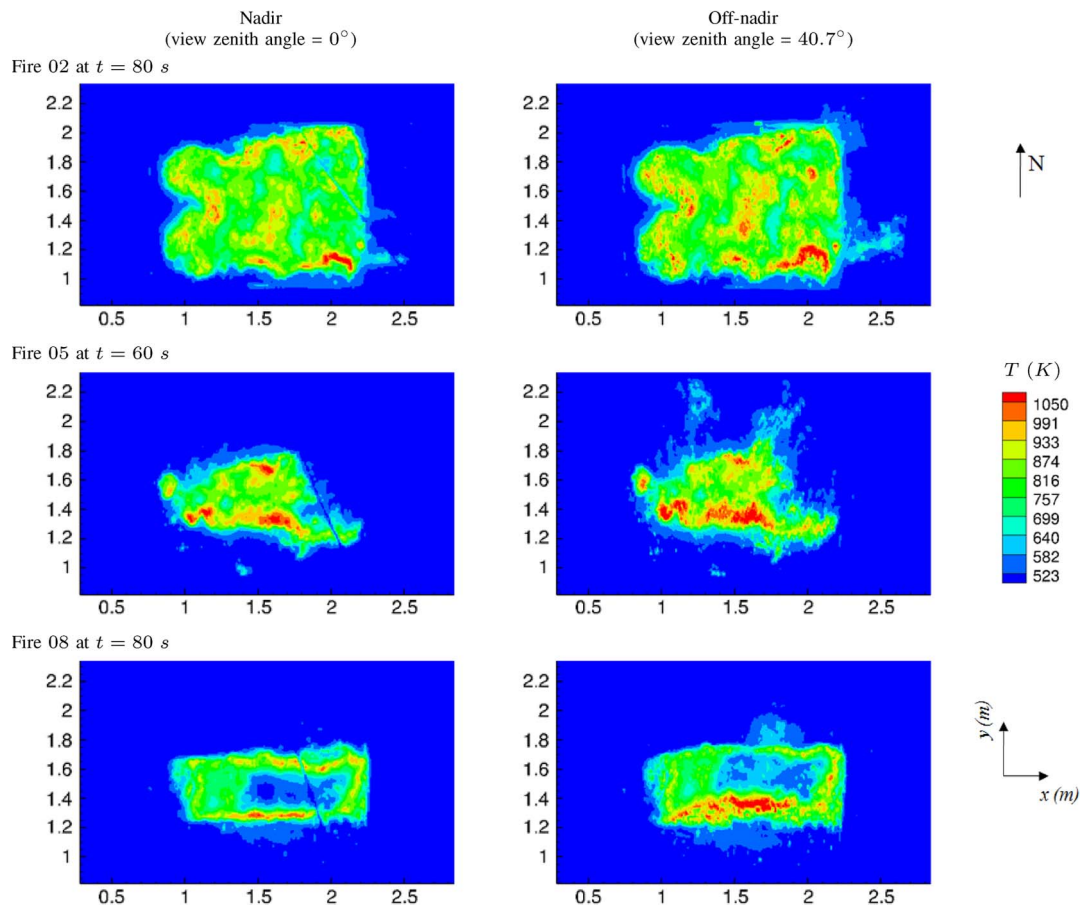


Fig. 5. Georeferenced BT images from the outdoor small-scale fire situation (Case study 2), shown for Fires 02, 05, and 08 collected during each fire's flaming phase from both the nadir and off-nadir viewing AGEMA 550 thermal imaging cameras. Fire characteristics are listed in Table I. Axis labels (x, y) are in meters, and for the sake of clarity, the y -direction is referenced as North.

FRE when viewed off-nadir. For example, Fire 03 has an off-nadir FRE more than 20% higher than for the nadir view, while Fire 07 (which only differed from Fire 03 by not having the wire mesh upon which the fuel bed rested and which therefore had a reduced 3-D structure) shows a FRE difference of only 10%. To further demonstrate that the difference in FRE between the nadir and off-nadir views is enhanced by an increased presence of flaming activity, Fig. 7 shows the correlation (r) between the warped nadir and off-nadir BT images, as well as the time series of FRP measured from nadir and off-nadir for Fires 02, 05, and 08. As already mentioned, the (assumed fixed) position of the cameras was slightly perturbed by the wind, and the off-nadir camera was particularly affected by this (as it was mounted on a weaker frame than was the nadir-viewing camera). Such movement resulted in error in the specified position of the GCPs and a reduced correlation coefficient (r) between the matching warped nadir and off-nadir images. In the case of Fire 02 for example, r falls below 0.8 for most of the fire duration. However, if we assume that perturbation by the wind causes some lateral shift in the image, then we can estimate a corrected correlation r_c by applying such a shift. The effects of camera rotation can be assumed negligible because (i) the perturbations are small (a few degrees of rotation only) and (ii) the DLT algorithm can correct most of the pixel size variation induced by small rotations (the distance between the camera and the fire is larger than the fire size, and so effects of camera

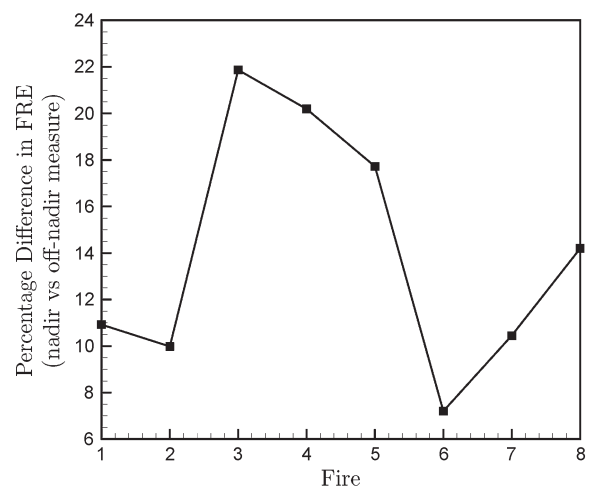


Fig. 6. FRE data from the outdoor small-scale fire situation (Case study 2), shown as the percentage difference in FRE as calculated from the nadir and off-nadir AGEMA 550 thermal imaging camera data of Fires 01 to 08, whose characteristics are listed in Table I. The off-nadir view angle was 40.7° . All measurements were made simultaneously from the two identical and intercalibrated thermal imaging cameras. FRE difference varied between 7% and 22% and was highest for the fires with greater flame height (see main text).

rotations are similar for every pixel, and this makes the linear correction embedded in the DLT algorithm appropriate for use in this situation). The corrected correlation r_c is defined as the maximum correlation between the matching nadir and off-nadir

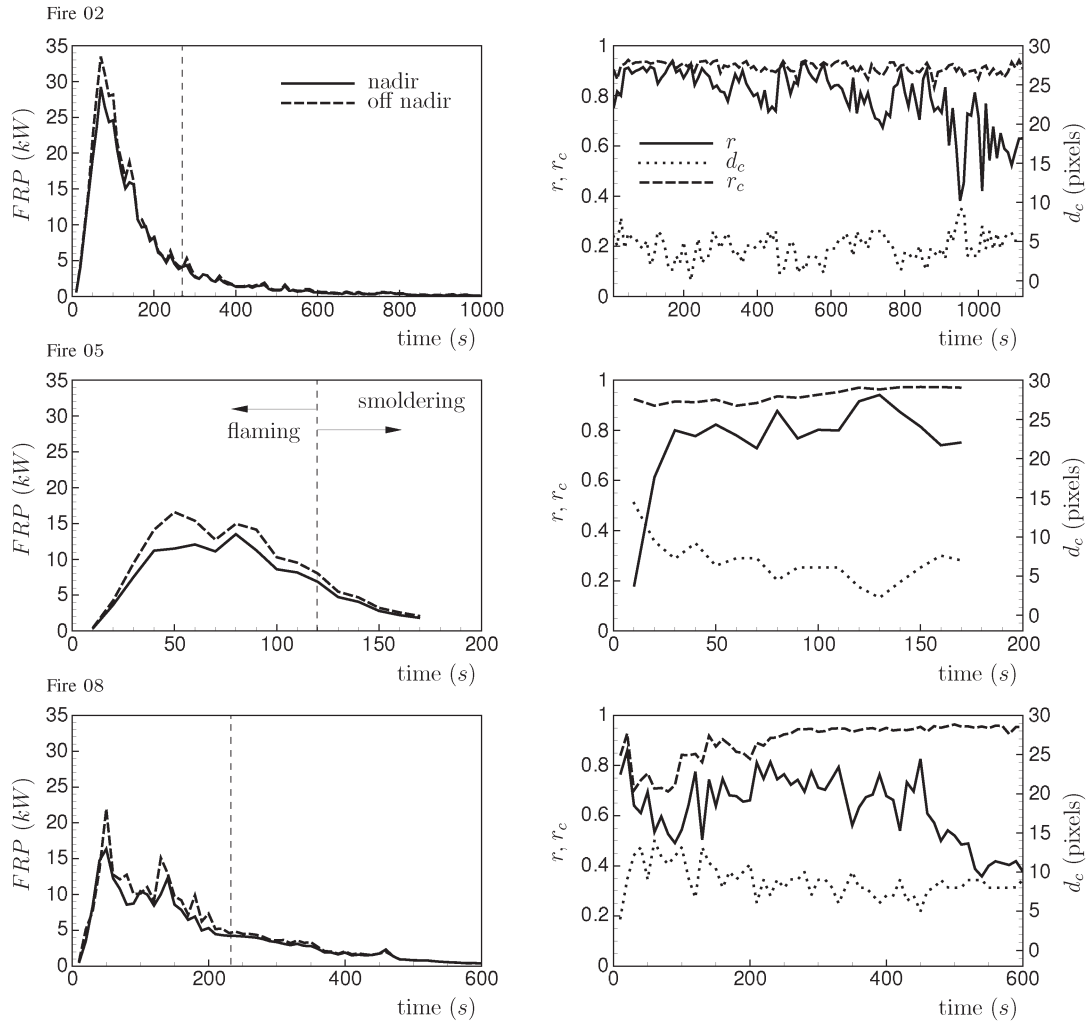


Fig. 7. Time series data for Fires 02, 05, and 08 of the outdoor small-scale fire situation (Case study 2). Shown at the left are the FRP measures calculated from the geometrically warped nadir and off-nadir viewing AGEMA 550 thermal imaging camera data, with the vertical dashed line indicating the time of transition between the fire's flaming and smoldering regimes (as deduced from an optical camera video record). At the right is shown the degree of spatial correlation in the pixel BT distribution between the warped nadir and off-nadir viewing thermal camera data. (Solid line) Initial correlation (r) is improved to (dashed line) the corrected correlation (r_c) by a variable spatial shift applied between the temporally matched nadir and off-nadir images. (Dotted line with right axis) Shift of d_c pixels is that which maximizes the value of r_c .

images when the off-nadir image is shifted by some pixels within a window of 20×20 pixels. The length of the shift vector d_c is also reported in Fig. 7 (right hand side of the y -axis), and for all fires, it remains below 15 pixels. This corrected correlation measure (r_c) indicates that, when the nadir versus off-nadir FRE difference is large (i.e., Fires 05 and 08 as shown in Fig. 7), the correlations are lower during the flaming regime than during the smoldering regime (the flaming and smoldering transition times reported on Fig. 7 were evaluated from the optical video camera record). We can therefore deduce from Fig. 7 that, when there is strong flaming activity, the measured FRP is more affected by the sensor view angle.

C. Case Study 3—Large Open Burn Plot

Here, we apply the complete methodology described in Section III (i.e., autocontrol point locating plus the DLT algorithm) to the data of a 945-m^2 experimental fire for which a time series of thermal images was collected using a handheld AGEMA 550 thermal imager operated from a hovering

helicopter. The fire was conducted in Northumberland (U.K.) in March 2010 [29] and was also observed by various remote sensing instruments housed on a Dornier 228 fixed-wing aircraft operated by the NERC Airborne Research and Survey Facility (ARSF; <http://arsf.nerc.ac.uk/>), including a high-spatial-resolution (39 megapixel) Leica RCD105 digital frame camera. The main aim of the thermal data processing procedure was to demonstrate the production of a time series of geolocated per-pixel BT and radiance measures, from which maps of FRP_p , FRE_p , and flame front ROS can be derived. As shown in Fig. 8, the burn plot was a rectangular patch of heather ($45\text{ m} \times 21\text{ m}$). *In situ* sampling was used to determine the fuel load ($1.5 \pm 0.5\text{ kg} \cdot \text{m}^{-2}$) and moisture ($60 \pm 2\%$), and a weather station and a 3-D sonic anemometer were used to record ambient weather conditions. The fire was ignited simultaneously on the northern plot edge as a back fire line and at the southeast corner as an ignition point. The back fire line ignition ended at $t = 53\text{ s}$, while the flaming combustion within the plot ceased at $t = 382\text{ s}$. The distance between the helicopter and the plot ranged from 130 to 170 m,

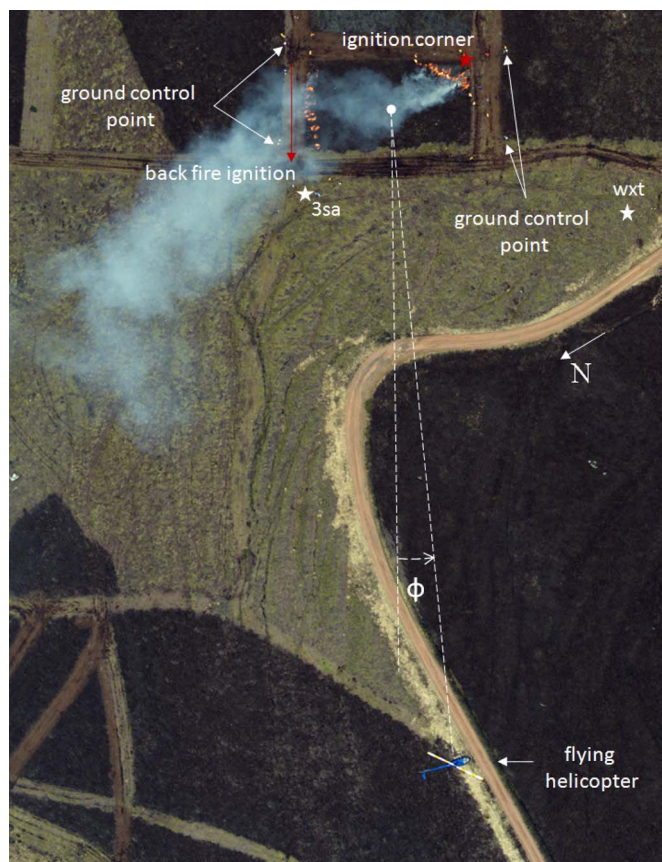


Fig. 8. Spatial layout of a large-scale (945 m²) experimental fire conducted in a U.K. heather moorland in March 2010 (Case study 3). The image was collected at $t = 78$ s after ignition by the Leica *RCD105* digital frame camera operated from the NERC ARSF Dornier 228 aircraft. Marked are the locations of the weather station (*wxt*), the 3-D sonic anemometer (*3sa*), the four thermal GCPs located at the plot corners (used in the DLT warping algorithm), and the helicopter from which the AGEMA 550 was operated. The helicopter was hovering around 113 m above the ground at this time and was observing the fire plot with view azimuth and zenith angles of around $\Phi = 8^\circ$ and $\theta = 55^\circ$, respectively. The view azimuth angle Φ is defined from the reference axis normal to the longest edge of the plot, as shown in the figure. The full time series of Φ and θ is reported in Fig. 11. The burning plot is 45 m long and 21 m wide and surrounded by a fire break of a few meters in width.

so according to Step 5 of our geometric correction methodology (see Section III), a resolution of 18 cm was selected for the final warped thermal imagery, which had a mean image frequency of 0.36 Hz. Fig. 9 shows the output FRE_p map calculated from the individual geometrically corrected FRP_p maps, along with snapshots showing the propagation of the fire front (the full animation can be seen at <http://wildfire.geog.kcl.ac.uk/shared/paugam-tgrs-2013.html>). The spatially integrated FRP time series for the entire plot is reported in Fig. 10(d), calculated from both the DLT and APS geometric correction algorithms. Although the two time series are similar and the final FRE difference between them is less than 1.5%, the APS method shows a noticeably higher FRP toward the end of the fire ($t > 260$ s). According to the ambient meteorological conditions [see Fig. 10(a)–(c)], there is no obvious reason for this reported increase in fire activity, and Fig. 9 shows that, after $t = 260$ s, when the back fire and the head fire connect, the active fire (AF) area is actually decreasing. In fact, the increase in reported FRP calculated via the APS method is

due to a significant shift in the helicopter position from which the thermal imagery was collected, as demonstrated in Fig. 11. Around $t = 240$ s, the hovering helicopter started to move from its initial location, which resulted in (i) an alteration of the viewing distance to the plot [Fig. 11(a)], and (ii) changes of a few degrees in both azimuth and vertical view angles [Fig. 11(b)]. Since the APS method only adjusts for variations in viewing distance and zenith angle, the resulting FRP measure can be affected by camera movement, whereas the DLT algorithm is able to better correct for these effects via the information obtained from the spatially invariant thermal GCPs. The DLT algorithm therefore adjusts the data for pixel size changes across the scene when warping to the base image frame, without requiring the exact knowledge of the camera position or orientation. The DLT algorithm also allows the calculation of the per-pixel FRE map from the geocoded FRP images, whereas the APS method provides only plot-integrated FRE. Fig. 10(d) also shows the time series of fire radiative flux (FRF, $\text{kW} \cdot \text{m}^{-2}$), calculated as the ratio of FRP and AF area, both derived from the results obtained with the full geometric correction algorithm. We defined AF area as the total surface area covered by pixels having BTs > 500 K. The FRF temporal evolution illustrates the different fire stages, first with a radiation intensity peak caused by the back fire (around $t = 40$ s), which is due to the combustion of the drip torch fuel used for igniting the back fire line ($t < 53$ s). It cannot be due to the front fire since, at that stage, the intensity of the fire front is still relatively weak as shown in the final FRE_p map of Fig. 9. Indeed, we have to wait until around $t = 100$ s to observe a high local FRE_p (> 0.09 MJ). This transition (at $t > 100$ s) can also be seen in Fig. 10(d), with a significant increase of FRF from 12 to 14 $\text{kW} \cdot \text{m}^{-2}$.

Further evaluation of the DLT algorithm output can be gained from the comparison of the thermal images taken from the helicopter and the high-spatial-resolution (5 cm) Leica digital frame camera imagery. To georeference these optical images, we used the same methodology as for the helicopter-borne thermal camera data (see Section III), using the same output spatial resolution as for the warped thermal imagery, i.e., 18 cm. Because of the fine spatial detail of the raw optical imagery, the pixel locations of the GCP fires were easily identifiable, even when partially covered by smoke. Fig. 12 shows the warped optical imagery collected at $t = 78$ and $t = 340$ s, overlain by the matching isocontour of thermal image BTs. The geometric match between the data appears good, with the $T > 700$ K isocontour derived from the warped thermal imagery correctly outlining the location of the flames evident in the optical imagery. The data of Fig. 12 also provide information useful for assessing the importance of flaming activity in FRP generation. At $t = 78$ s, the fire is still influenced by the ignition pattern, so we only base our discussion on the $t = 340$ s data. Assuming that, at this high (18 cm) spatial resolution, the ground pixels can be considered thermally homogeneous, an analysis is carried out on the contribution of smoldering and flaming combustion to the measured FRP. Burned areas located immediately behind the fire front have BTs of $500 \text{ K} < T < 600 \text{ K}$ and cover approximately the same spatial extent as the flaming zone (illustrated by the isocontour $T > 700$ K).

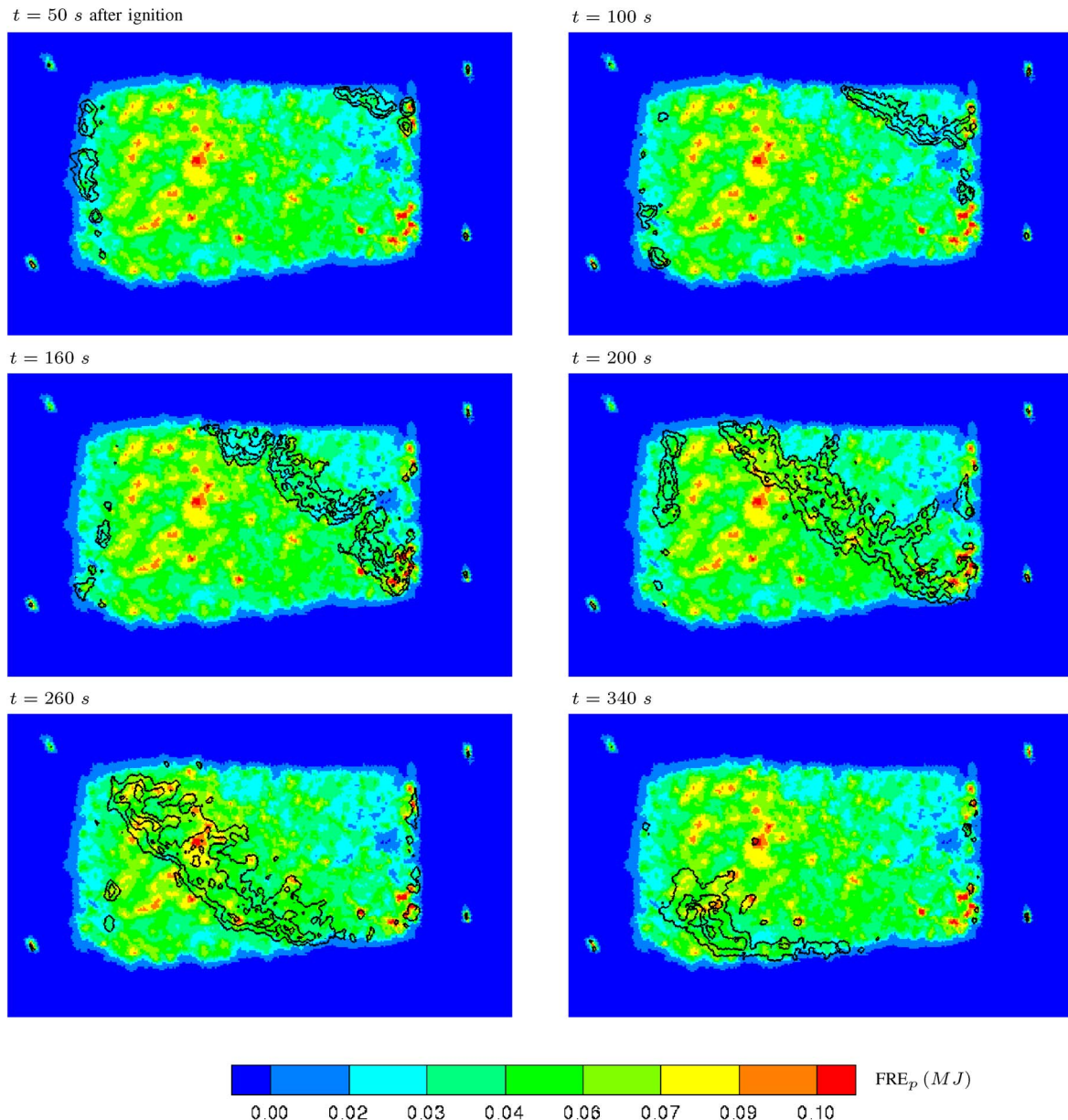


Fig. 9. Thermal camera data from the outdoor large-scale ($45 \text{ m} \times 21 \text{ m}$) heather moorland experimental fire shown in Fig. 8. These data were derived from imagery collected by the off-nadir viewing handheld AGEMA 550 used from the helicopter shown in that figure and are processed via the DLT geometric correction algorithm. Shown are the fire front location at $t = 50$ s to $t = 340$ s as depicted by black lines indicating where local BTs exceed 600 K. The background shows the final spatial map of FRE_p , calculated from the temporal integration of the individual FRP_p maps derived at the full base map spatial resolution output from the DLT algorithm.

Using the Stefan–Boltzmann equation ($\propto T^4$), we therefore estimate that the cooling area contributes at least four times less FRP than the flaming zone (i.e., $(500/700)^4 \sim 0.25$). Thus, Fig. 12 indicates that the dominant source of emitted radiation is the pixels in the flaming zone, suggesting that assumptions of the MIR radiance method, which is focused on optimizing the accuracy of FRP retrieval from areas of active flaming or smoldering combustion (whose temperature is assumed to exceed 700 K) and which tends to underestimate FRP from cooler surfaces, are well founded [21], [25]. Another feature of Fig. 12 is related to the flaming area ($T > 700$ K), where, on the left side of the image, ahead of the fire front, there is a part of the fuel bed which appears neither burned nor covered by flame but which seems to be contributing a high FRP (i.e., lies

within the red thermal contour). The thermal radiation detected by the AGEMA 550 from these apparently non-burning pixels is not produced by combustion occurring at this location, but rather could be due to (i) thermal radiation emanating from neighbouring (burning) pixels that is being absorbed and re-emitted (and also potentially directly reflected) by this area, or perhaps more likely to (ii) hot soot particles within the flame and the geometric distortion effects related to flame height originally highlighted in Section III and which were seen clearly in the previous smaller-scale case studies. In case (i), this radiation has already been accounted for via the assumption of isotropic emission characteristics and the semihemispherical integration of the directionally measured radiance in the FRP calculation [21], and in case (ii), we are potentially dealing with

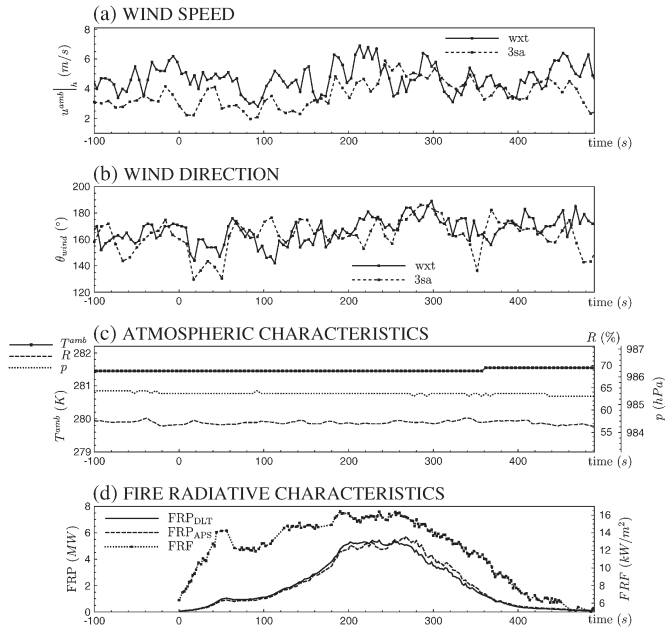


Fig. 10. Time series measurements from the large-scale experimental heather moorland fire and shown in Figs. 8 and 9. Panels (a) and (b) show horizontal wind speed ($u^{\text{amb}}|_h$) and direction (θ_{wind}) as measured by the *wxt* weather station and the *3sa* 3-D sonic anemometer whose location relative to the burning plot was shown in Fig. 8. Panel (c) shows the atmospheric pressure (p), relative humidity (R), and temperature (T^{amb}) as measured by the *wxt* weather station. Panel (d) shows the FRP retrieved from the helicopter-borne AGEMA 550 imagery, as calculated via both the DLT and APS algorithms described in Section III, together with the FRF derived from the FRP_{DLT} measure only. Note: On the lower two panels, the symbols marking the measurement time are only reported on T^{amb} and FRF time series, in order not to crowd the figure. Measurement times for the other variables are identical to these.

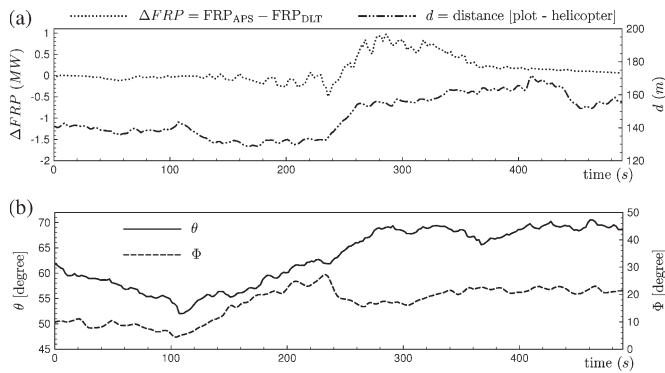


Fig. 11. (a) Difference in FRP as retrieved from the DLT and APS methods and calculated from the helicopter-borne AGEMA 550 imagery of the large-scale (945 m²) experimental fire (Case study 3) shown in Figs. 8 and 9. The distance between the plot center and the helicopter (d_h) is also shown in (a), while the original FRP data calculated using both approaches were shown in Fig. 10(d). (b) Azimuth (Φ) and zenith (θ) angles between the AGEMA 550 and the plot center (whose definition is shown in Fig. 8).

narrow-band radiation initially emitted by heated gases within the flame, but which is then absorbed by the soot particles and reemitted as Planckian radiation (including at the measured MIR wavelength); potentially, both these situations can lead to FRP overestimation. The question is then to evaluate how much of the radiant heat produced in the combustion zone is directly related to combustion and how much of that is directly related to heat transfer from neighboring pixels or to geometric effects induced by soot radiation. FRP and FRE have been designed as

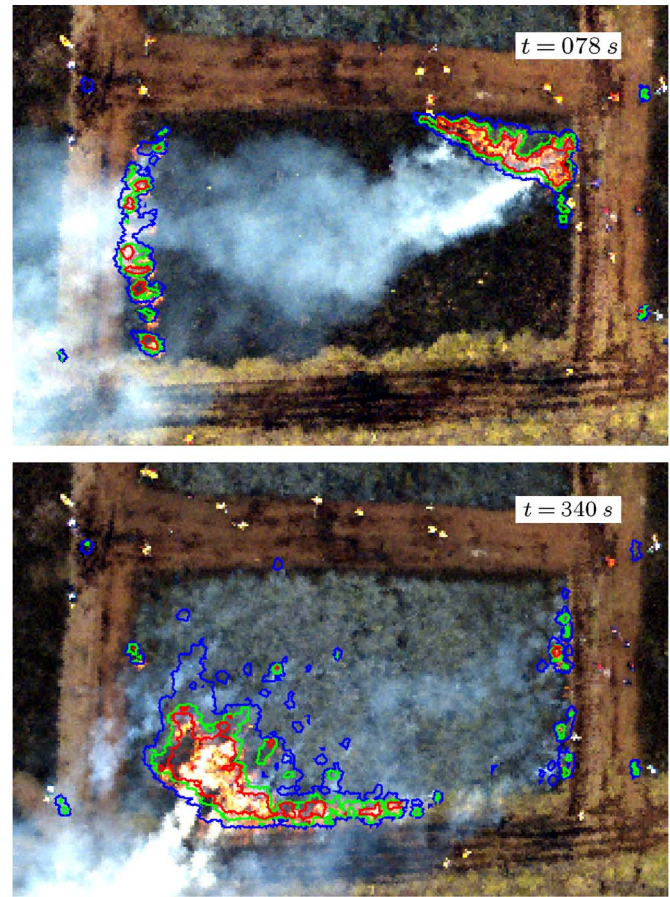


Fig. 12. True color images overlain by isocontours of fire pixel infrared BT calculated from the helicopter-borne AGEMA 550 thermal camera imagery of the outdoor large-scale experimental fire (Case study 3) shown in Fig. 8, where the imagery has been warped using the DLT algorithm (500 K isocontour is shown as blue, 600 K as green, and 700 K as red). Shown are the situations at (top) $t = 78$ s and (bottom) $t = 340$ s. The true color images come from the simultaneously acquired Leica RCD105 high-resolution digital frame camera data taken by the NERC ARSF aircraft. A good spatial match between the locations of the highest temperature areas and the positions of the flames and smoke plume source regions is clearly apparent. The size of the fire plot is 45 m $\times$ 21 m, and the yellow points around the perimeter of the burned plot are fire-service personnel (for scale).

measures ideally proportional to fuel consumption [21], [25] in order that their assessment enables the estimation of the resultant fire trace gas and aerosol emissions (e.g., [10], [12], [13], [22], and [30]). The detection of MIR radiation that is only indirectly produced by combustion, i.e., via combustion radiation reflection (or absorption and reemission), may affect the accuracy to which FRP can be measured, which, in turn, has implications for closure of the fire energy balance [31]. Future studies at the scale of experimental fire plots, together with numerical simulations using physically based fire models (e.g., [26]), may assist in better understanding the significance of these effects. Despite the presence of the occasional nonuniformity highlighted earlier, in general, the positions of the high-temperature thermal features and the optically imaged flames shown in Fig. 12 appear very well matched, and indeed, we observe a strong correspondence between the time-integrated FRE_p map and the postfire aerial photography. In particular, Fig. 13 highlights parts of the plot where there are particular heterogeneities (i.e., gaps) in the FRE_p map and which match

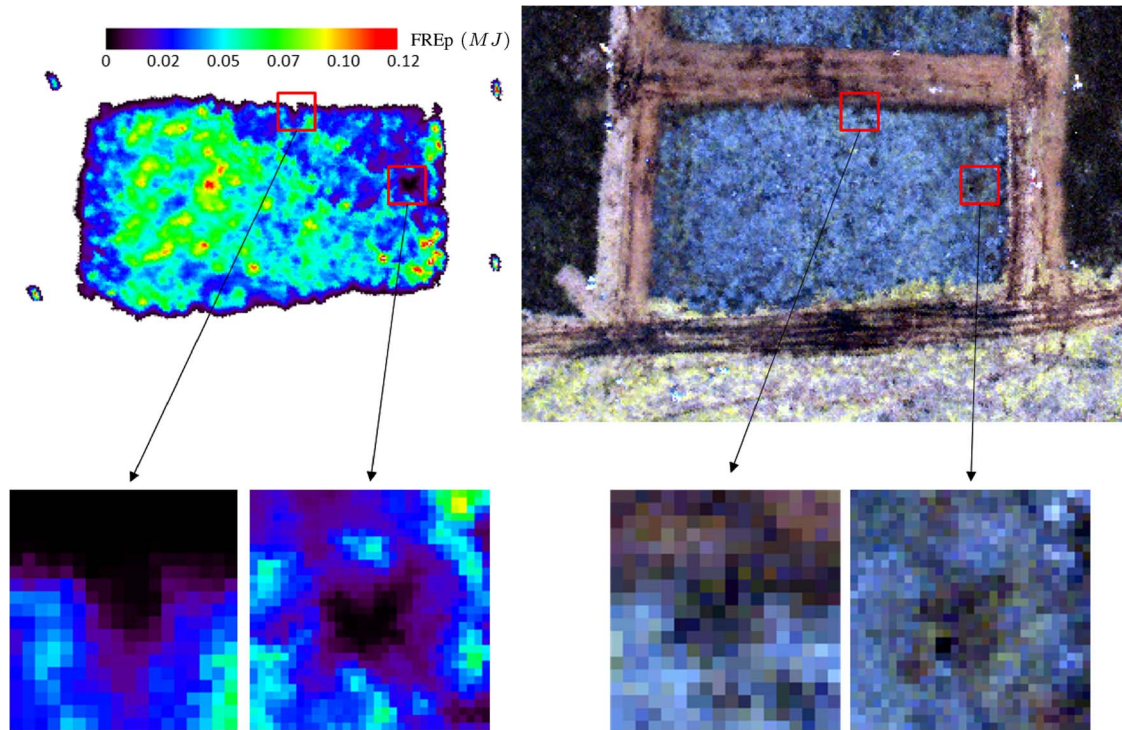


Fig. 13. Comparison of the FRE released per-pixel (FRE_p) map derived from the warped helicopter-borne AGEMA 550 thermal camera imagery of the large-scale experimental heather moorland fire (Case study 3; shown in Fig. 8) and the postfire image collected by the airborne Leica RCD105 high resolution digital frame camera, where burned areas (ash) appear light and unburned areas of heather appear dark in the true color imagery. Lower panels depict subregions where similar patterns can be seen in the FRE_p and true color imagery (i.e., areas of zero FRE_p matching to areas of unburned fuel). Each pixel in the FRE_p map corresponds to an area of $18\text{ cm} \times 18\text{ cm}$, and the total fire plot size is $45\text{ m} \times 21\text{ m}$.

to apparently unburned fuel patches detectable in the postfire aerial photography. This demonstrates that, at the 18-cm output spatial resolution used here, the FRE_p measure appears to provide a good metric with which to map spatial variations in fire behavior. More detailed comparisons quantitatively relating spatial maps of FRE_p and maps of fuel consumption at the types of fire illustrated here will be the subject of future work.

For this fire, the difference between the FRE calculated using our DLT-based approach and the APS method is on the order of a few percent [see Fig. 10(d)]. This small difference is influenced by the fact that the plot is spatially relatively modest in area, so the AGEMA 550 with the 20° lens viewed all points of the fire scene at a similar viewing angle and distance. In case of larger plots, for example, the 7-ha Kruger fire plots whose thermal imaging is described in [22], the helicopter would typically either have to hover at a higher altitude, which can make the measurement more difficult as the helicopter can have problems maintaining a relatively fixed position, and/or use a wider angle lens, which would cause the pixel sizes across the plot to vary more widely. This would reduce the validity of the APS method assumptions.

V. ROS

As the final component of this study, we demonstrate the relevance of the developed georeferencing methodology to the derivation of flame front ROS. Our target is the 945-m^2 heather fire detailed in Section IV-C. To our knowledge, this is the first attempt to derive an accurately mapped representation of flame front ROS variations using imagery recorded with a handheld

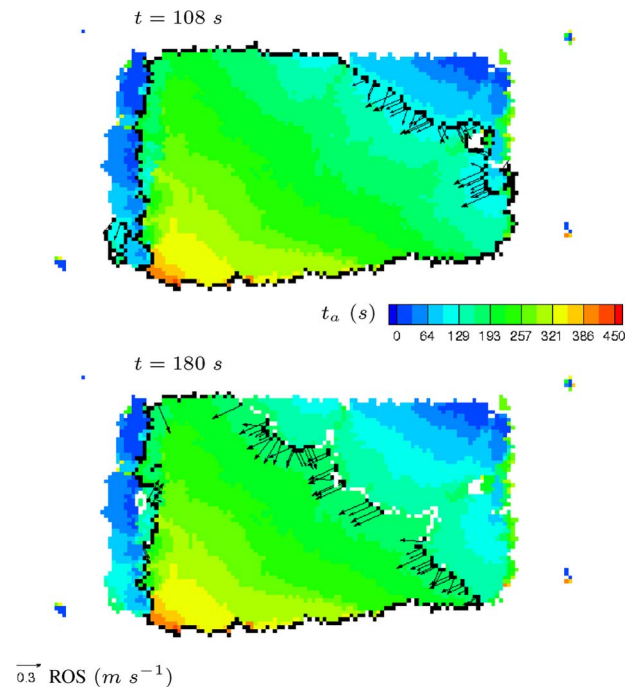


Fig. 14. Map depicting the fire front arrival time (t_a) for the large-scale ($45\text{ m} \times 21\text{ m}$) experimental heather fire (Case study 3; shown in Fig. 8) discussed in Section IV-C. The fire front arrival time is defined in seconds after fire ignition and is calculated on a per-pixel basis as the time when the pixel infrared BT first exceeds 600 K . The black line depicts the fire front position extracted from the warped AGEMA 550 thermal imagery at (left) $t = 108\text{ s}$ and (right) $t = 180\text{ s}$. The black arrows represent the fire front ROS at these same times, calculated as described in Section V. The white pixels represent areas of the fire front where the ROS was unable to be determined.

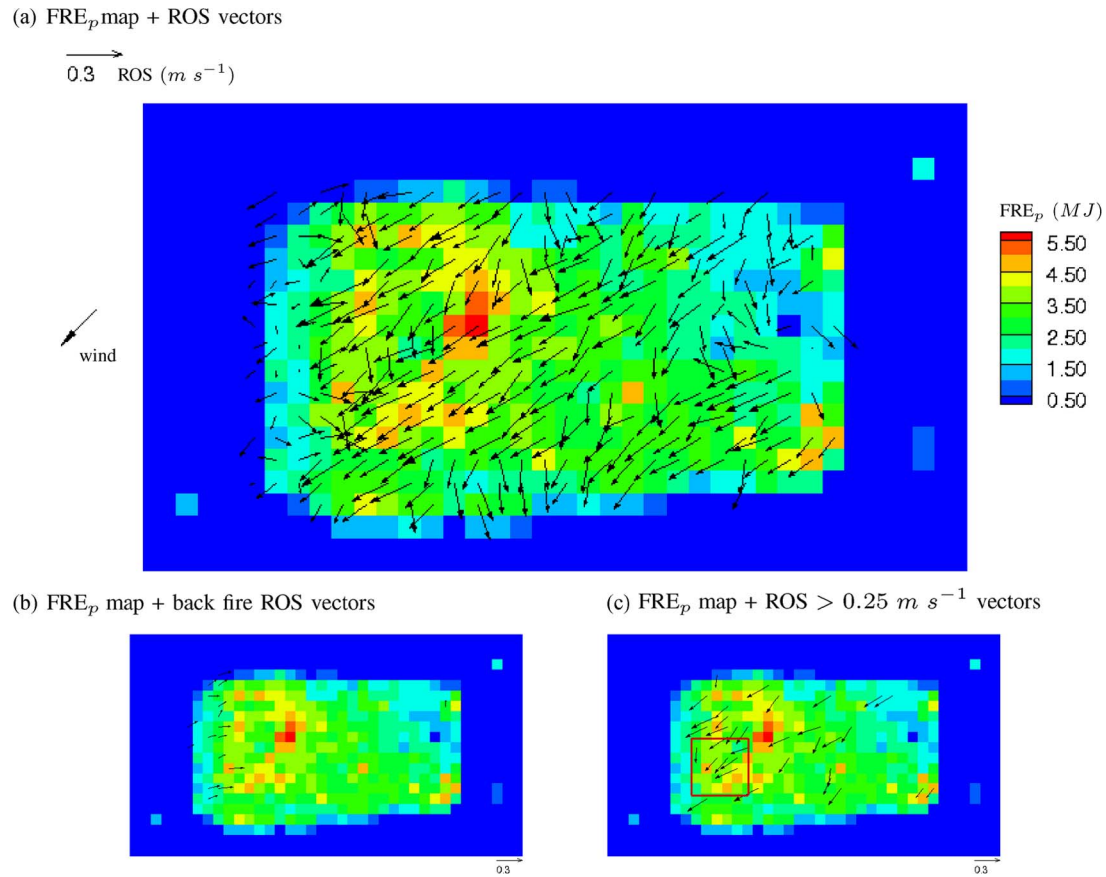


Fig. 15. Vectors depicting the spatial variation in flame front ROS for the large-scale ($45 m \times 21 m$) experimental heather moorland fire (Case study 3; shown in Fig. 8). The ROS data shown here were calculated over clusters of 8×8 pixels in order to reduce the number of pixels for which ROS remained undetermined, as described in Section V. The ROS vectors are overlain on a map of FRE released at each pixel (FRe_p) and displayed at the same reduced spatial resolution ($\sim 2 m^2$ per pixel). Panel (a) shows the full plot, while panels (b) and (c) highlight respectively the back fire spreading upwind and the locations where ROS exceeds $0.25 m s^{-1}$.

thermal camera operated from a helicopter. Previous studies have either used fixed-wing airborne sensors (e.g., [4] and [5]), allowing only a lower temporal resolution investigation of the intrafire variability, or been limited to a more constrained experimental setup, such as in [17] where a vertically viewing helicopter platform was used with no need for view angle corrections. The analysis is only possible with the set of warped imagery provided by the DLT method but not by the APS approach. The fire front arrival time t_a at a pixel is defined as the time when its georeferenced BT first exceeds 600 K (see Fig. 13). This temperature corresponds to the start of fuel vaporization in the pyrolysis processes. Then, at a time t during the fire propagation, the set of pixels belonging to the fire front is determined from the isoline of the arrival time map, and the ROS of each pixel is calculated following a similar approach as that in [6]. Assuming that a moving fire front is developing as an approximately elliptical shape [32], the fire is expected to propagate in the horizontal direction normal to its surface. Thus, the intersection of the normal and the fire front at the time $t + dt$ (i.e., the time of the next available image) gives the distance covered during the period dt , and thus, the local ROS can be calculated. In reality, the fire propagation can be highly variable, and the derivation of the fire front surface normal can be difficult. To simplify this step, the initial 18-cm spatial resolution of the georeferenced imagery was decreased

by a factor of 2 to reduce the effect of such variations. The fire front normal was then derived from the perpendicular to the line formed by the two closest AF pixels. The vector field in Fig. 14 shows the derived local fire front ROS, at $t = 108 s$ and $t = 180 s$ after fire ignition. In Fig. 14, pixels coded as white indicate locations where the ROS could not be determined, due to (i) unburned area such as those highlighted in Fig. 13 (and which are present here on the right of the arrival time map as they never experience temperature $> 600 K$), or (ii) to the complex geometry of the fire front around time t , which prevented definition of the intersection between its normal and the fire front at time $t + dt$. Nevertheless, despite the relatively simple method used to derive the surface normal for each fire front location, the ROS was able to be evaluated for more than half of the pixels in the burn plot over the entire fire duration. Based on the assumption that the ROS was relatively homogeneous at the $\sim$ meter scale, a map of ROS (see Fig. 15) was derived by averaging the per-pixel ROS estimates over clusters of 8×8 pixels (i.e., resulting in a pixel spatial resolution of 1.44 m) and considering the cluster ROS to be invalid when less than 20% of its surface had a retrieved ROS value (see white pixels of Fig. 16). The maps of Fig. 15 identify where back fire activity occurred [i.e., fire propagating upwind; see Fig. 15(b)] and where the fire front accelerated [see Fig. 15(c)]. The maximum local ROS identified is $0.36 m s^{-1}$, and the mean

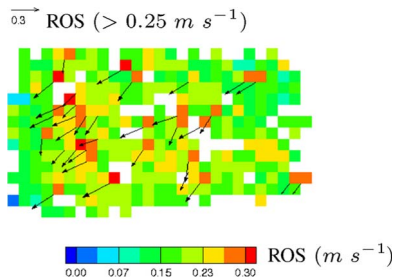


Fig. 16. Map of flame front ROS for the outdoor large-scale experimental heather fire described herein (Case study 3; shown in Fig. 8 and in Fig. 15). The overlain vectors depict the point where $ROS > 0.25 \text{ ms}^{-1}$, areas previously depicted in Fig. 15(c).

value for the plot is 0.19 ms^{-1} . These values agree relatively well with the previous study in [33], conducted in a similar mature heather fuel type and under similar meteorological conditions. In that work, the authors derived ROS by timing the fire front propagation between two posts located 5 m apart. With a similar wind speed as the one experienced in the current case study [5 ms^{-1} ; see Fig. 10(a)], Davies *et al.* [33] measured a fire front ROS of 0.14 ms^{-1} , and their measurement approach is less likely to capture local accelerations of the fire front which our spatial mapping can more completely record. Mapping the ROS also illustrates the effect of the back fire on the main head fire's propagation, with a visible acceleration in the frontal velocity around the location where the back fire and head fire connected (i.e., a high ROS toward the left of Fig. 16; also see the rapid movement of the head fire front shown in Fig. 9 between $t = 200 \text{ s}$ and $t = 260 \text{ s}$). This acceleration seems likely to be due to a combination of (i) increased pre-heating of the fuel due to the presence of the back fire, and/or (ii) a stronger local wind generated by the updraft of the back fire. However, we also note that this area of acceleration seems unconnected with a high FRE value. Since FRE is correlated to fuel consumption [21], the low FRE_p and high ROS in the boxed area of Fig. 15(c) indicate the fire to have experienced an area of fast propagation over a relatively wide zone (as indicated by the converging high ROS stream lines) where it consumed less fuel per unit area compared to the immediately surrounding (yellow orange) regions. Potentially, at this location, the lower amount of fuel required less energy for preheating and pyrolysis and, therefore, less time to reach the detection threshold of 600 K. Conversely, a high ROS is not always equivalent to a low FRE [e.g., in the middle and right-hand regions of Fig. 15(c)]. Rather, fuel consumption (and thus FRE) at a "point" is related more to the fire's residence time than to its ROS (e.g., [34]). To go further and gain a better understanding of the fire behavior through the use of these data requires detailed information on fuel consumption, a task that will be tackled in future work.

VI. CONCLUSION

In this paper, we have demonstrated the development, testing, and verification of algorithms to derive georeferenced maps of fire pixel BT, radiance, and FRP from the data collected by nadir or off-nadir viewing thermal imagers viewing small-to large-scale vegetation fires. The ultimate focus of the work has been to use these methodologies to derive spatially ex-

plicit information on FRP, FRE, and fire front ROS. We have demonstrated the technique's performance with data from a small ($< 2 \text{ kg}$) thermal imaging camera operating at MIR wavelengths and which we have used to collect data of various types of "fire scene" in different measurement modes: tripod mounted, from an elevated platform, and in handheld mode from a helicopter. The spatially explicit data on FRP, FRE, and ROS captured by this remote sensing approach provide more information with regard to the variability of fire energy release and fire front behavior than do the more commonly used point-based approaches long employed by the fire science community (e.g., [3] and [35]). In particular, the spatially explicit maps of FRP provide the detail necessary to start to (i) study the effects of varying sensor view angle, and (ii) address the issue (when coupled with visible imagery) of the impact of direct v.s. indirect flame radiation. They also allow the derived maps of FRE to be related to the spatial variability of fuel consumption [12], [21]. In terms of ROS, our method allows this important parameter to be mapped and its variability across the burning plot to be examined and related to driving parameters such as wind speed, fuel availability, and the coalescing of separate fire fronts. Future work will utilize these methods to explore these types of relationships in more detail and will compare the measurements possible when using these techniques to matching numerical simulations from a physically based fire model parameterized with the same meteorological and fuel conditions.

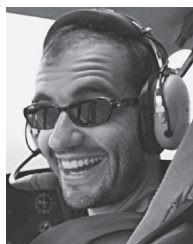
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